FortiAIGate Deployment

▼ 1. Environment Preparation and Validation

FortiAIGate installation must be performed from a system that has administrative access to the Kubernetes cluster.

▼ 1.1 Prerequisites

Ensure the following components are available before deployment:

- **Kubernetes 1.25.0 or later** (linux/amd64 or linux/arm64)
- **CNI Plugin** - Container Network Interface plugin installed and configured
  - Common options: Calico, Flannel, Weave Net, Cilium
  - Must support Pod-to-Pod communication
- **kubectl** - Ensure your client can access the Kubernetes API server
  - kubectl documentation
- **Helm 3.10.0 or later**
  - Helm installation guide
- Container registry ready and accessible
- Ingress Controller deployed
- Optional: GPU nodes (if GPU mode is required)
  - The supported GPU models include `NVIDIA L4`, `NVIDIA A10`, and `NVIDIA A100`.

▼ 1.2 Kubernetes Basic Verification

```
# 1. Check cluster connectivity
kubectl cluster-info
kubectl get nodes

# 2. Verify CNI plugin is working (kubeadm cluster)
kubectl get pods -n kube-system | grep -E "(calico|flannel|we
ave|cilium)"

# If you are using AWS EKS, please verify AWS VPC CNI
kubectl get ds -n kube-system aws-node
kubectl get pods -n kube-system | grep aws-node

# 3. Test Pod-to-Pod networking
```

```
kubectl run test-pod --image=busybox:1.35 --rm -it --restart=
Never -- sh -c "nslookup kubernetes.default.svc.cluster.loca
l"

# 4. Test DNS resolution
kubectl run test-dns --image=busybox:1.35 --rm -it --restart=
Never -- sh -c "nslookup google.com"

# 5. Verify Helm installation
helm version
```

- All nodes must be in `Ready` state.
- The CNI plugin (Calico / Flannel / Cilium / etc.) must be in `Running` state.
- Helm must be installed.

▼ 1.3 Ingress Verification

FortiAIGate requires a functional Ingress Controller.

1. **Check IngressClass**

   ```
   kubectl get ingressclass
   ```

   Example:

   ```
   NAME      CONTROLLER
   nginx     k8s.io/ingress-nginx
   alb       ingress.k8s.aws/alb
   ```

   Record the `NAME` value for ingress configuration in values.yaml.

2. **Verify Ingress Controller Pods (If using an ingress controller)**

   ```
   kubectl get pods -A | grep -i ingress
   ```

   Ingress controller pods must be `Running`.

▼ 1.4 Storage Verification

FortiAIGate requires persistent storage.

Before deployment, verify that your cluster has a valid StorageClass or available PersistentVolumes (if static provisioning is used).

1. **Verify StorageClass**

   List available StorageClasses

```
kubectl get storageclass
```

Example output:

```
NAME              PROVISIONER           DEFAULT
gp3 (default)     ebs.csi.aws.com       Yes
nfs-storage       nfs.csi.k8s.io
```

What to Check

- Identify whether a StorageClass is configured
- The storage type should support `RWX` (ReadWriteMany).
  - AWS EBS (gp2/gp3) supports `RWO` only. Use a shared storage solution (e.g., Amazon EFS) for `RWX` access.
- Record the StorageClass name for storage configuration in values.yaml.

2. **Verify PersistentVolumes (PV)**

   List PersistentVolumes:

```
kubectl get pv
```

   Example:

```
NAME        CAPACITY   ACCESS MODES   RECLAIM POLICY   STA
TUS
pv-nfs-01   100Gi      RWX            Retain           Ava
ilable
```

   What to Check

   - PVs should be in `Available` state
   - Capacity should meet deployment requirements
   - AccessMode should match expected usage RWX (ReadWriteMany)
   - Record the PV name for storage configuration in values.yaml.

   If using dynamic provisioning (most common in cloud environments), PVs may not appear until a PVC is created. This is normal.

▼ 2. Image Preparation

▼ 2.1 Prerequisites

- Registry server is ready.
  - Since the images are OCI-based, the registry should support the OCI image specification and Docker Registry HTTP API v2.
- FortiAIGate images and Helm chart package received from Fortinet
  - Please download the latest build images from https://info.fortinet.com/builds/?project_id=807 （The latest image currently available on Infosite is build 0005, which is not our most recent build. We will continue updating the versions on Infosite, please stay tuned.)
- Docker installed
  - This guide uses the Docker CLI to push images to the registry, and a Linux system is recommended.

▼ 2.2 Load Images

```
sudo docker load -i <path_to_image>
```

Example:

```
sudo docker load -i FAIG_api-V8.0.0-build0004-FORTINET.tar
```

▼ 2.3 Tag Images for Your Registry

List images:

```
docker images
```

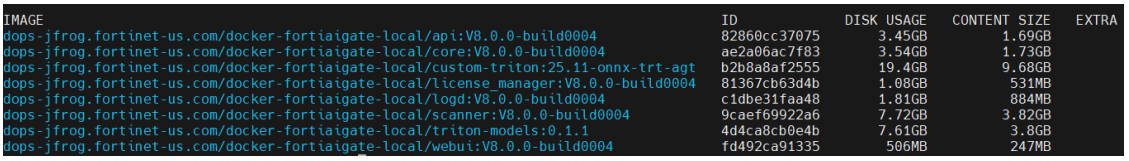


IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/api:V8.0.0-build0004	82860cc37075	3.45GB	1.69GB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/core:V8.0.0-build0004	ae2a06ac7f83	3.54GB	1.73GB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/custom-triton:25.11-onnx-trt-agt	b2b8a8af2555	19.4GB	9.68GB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/license_manager:V8.0.0-build0004	81367cb63d4b	1.08GB	531MB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/logd:V8.0.0-build0004	c1dbe31faa48	1.81GB	884MB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/scanner:V8.0.0-build0004	9caef69922a6	7.72GB	3.82GB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/triton-models:0.1.1	4d4ca8cb0e4b	7.61GB	3.8GB	
dops-jfrog.fortinet-us.com/docker-fortiaigate-local/webui:V8.0.0-build0004	fd492ca91335	506MB	247MB	

Tag images:

```
docker tag <source_image> <target_registry>/<repository>:<tag
>
```

Example:

```
docker dops-jfrog.fortinet-us.com/docker-fortiaigate-local/ap
i:V8.0.0-build0004 registry.example.com/api:build0004
```

⚠️ Keep original tags for:

- `triton-models`
- `custom-triton`

▼ 2.4 Push Images

If the target registry requires authentication, log in before pushing images:

```
docker login <registry_address>
```

If using AWS ECR, the login method is as follows:

```
aws ecr get-login-password --region <aws_region> \
| docker login --username AWS --password-stdin <ecr_registry>
```

Push image to registry server.

```
docker push <target_registry>/<repository>:<tag>
```

Repeat for all required images.

▼ 3. Helm Chart Configuration

Upload Helm Chart to the system that has administrative access to the Kubernetes cluster, and extract it.

▼ 3.1 Configure values.yaml

Edit `<helm_chart_extract_path>/fortiaigate/values.yaml` and update the parameter configurations to match your environment.

Key configuration:

1. **Image and GPU Configuration**

```
# FortiAIGate Global Configuration
fortiaigate:
  # Common image configuration
  image:
    repository: <your registry>
    tag: <image_tag>
    pullPolicy: IfNotPresent
    pullSecrets: []  # Configure this field when pulling i
mages from a private registry that requires authenticatio
n.

  # Global GPU configuration
```

```
  gpu:
    enabled: true  # Set to false if no GPU node
```

Note: FortiAIGate integrates with **NVIDIA Triton Inference Server** to provide AI model inference services. If no GPU is available and the system runs on CPU only, inference performance will be significantly reduced and may not meet production performance requirements.

2. Storage Configuration

Storage deployment modes:

1. Dynamic provisioning (default): Leave existingPvName empty
   - Creates PVC with storage class, auto-creates PV via storage class
2. Bind to existing PV: Set existingPvName to the name of existing PV
   - Creates PVC that binds to the specified existing PV
   - User must manually create the PV first

```
# Storage Configuration
storage:
  # Storage size and class configuration for API and Core
services
  size: 8Gi
  storageClass: ""  # Set to a storage class that exists i
n your environment. if empty, default storage class will b
e used

  # Use existing PV name (leave empty for auto-creation)
  # When specified, creates a PVC that binds to the existi
ng PV
  existingPvName: ""
```

3. Ingress Congiguration

```
ingress:
  className: "nginx" # Ingress class name. Must match your
installed ingress controller.
  host: ""  # Optional: Set to a specific hostname to enab
le host-based routing
  # Custom annotations that take priority over predefined
controller settings
  # When provided, these annotations will be used instead
of controller-specific ones
  #
```

```
  # NOTE: For NGINX and ALB ingress controllers, default T
LS backend annotations are automatically configured. Howev
er, if you're using a different ingress controller (Contou
r, Istio, etc.) and you enable TLS for backend, you MUST a
dd the appropriate annotations here to ensure HTTPS backen
d connectivity.
  #
  # Example for custom NGINX TLS backend (overrides defaul
t NGINX configuration):
  # annotations:
  #   nginx.ingress.kubernetes.io/backend-protocol: "HTTP
S"
  #   nginx.ingress.kubernetes.io/ssl-redirect: "true"
  annotations: {}
```

If you are using AWS EKS and ALB, please refer to the following ingress configuration.

```
ingress:
  className: "alb" # You need update this to your ingress
used
  host: "<your_domain_name>"
  annotations:
    kubernetes.io/ingress.class: alb
    alb.ingress.kubernetes.io/scheme: internet-facing
    alb.ingress.kubernetes.io/target-type: ip
    alb.ingress.kubernetes.io/backend-protocol: HTTPS
    alb.ingress.kubernetes.io/healthcheck-protocol: HTTPS
    alb.ingress.kubernetes.io/listen-ports: '[{"HTTP":80},
{"HTTPS":443}]'
    alb.ingress.kubernetes.io/certificate-arn: <ACM_cert_a
rn>
    alb.ingress.kubernetes.io/ssl-redirect: '443'
```

4. TLS Congiguration

Upload your certificate and key to `<helm_chart_extract_path>/fortiaigate/files/certificate`, and then configure the following parameters.

```
# TLS Configuration
tls:
```

```
  # Enable/disable TLS for all services except postgresq
l and redis
  enabled: true

  # TLS mount path in containers
  mountPath: "/etc/tls"

  # Certificate and key (used to create secrets) for For
tiAIGate TLS services
  # You need to provide your own certificate and key
  cert: files/certificate/dflt.crt
  key: files/certificate/dflt.key
```

▼ 3.2 Parameter Reference

Global

Name	Description	Default Value
global.security.allowInsecureImages	Allows pulling container images without verification.	true
global.licenses	Map of node names to license files. Pods are scheduled only on these nodes. Configuration required.	{}

FortiAIGate Global Configuration

Name	Description	Default Value
fortiaigate.image.repository	Docker image repository	
fortiaigate.image.tag	Image tag	
fortiaigate.image.pullPolicy	Image pull policy	IfNotPresent
fortiaigate.image.pullSecrets	Image pull secrets	[]
fortiaigate.gpu.enabled	Enables GPU scheduling for FortiAIGate pods that require GPU resources. Set to false to run entirely on CPU nodes	true
fortiaigate.env.logLevel	Global log level	"INFO"

API Service

Name	Description	Default Value
api.replicas	Number of API replicas	1
api.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed	500m

Name	Description	Default Value
	this limit.	
api.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	512Mi
api.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	250m
api.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	256Mi

Web UI

Name	Description	Default Value
webui.replicas	Number of Web UI replicas	1
webui.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	500m
webui.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	512Mi
webui.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	250m
webui.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	256Mi

Core Service

Name	Description	Default Value
core.replicas	Number of Core replicas	1
core.env.maxParallelRequests	Maximum number of concurrent requests processed by Core	100
core.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	1000m
core.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	4Gi
core.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at	1000m

Name	Description	Default Value
	least this amount.	
core.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	2Gi

Triton Inference Server

Name	Description	Default Value
triton.replicas	Number of Triton replicas	1
triton.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	2000m
triton.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	20Gi
triton.resources.limits.gpuCount	Number of GPUs	1
triton.resources.limits.gpuResourceName	GPU resource name	"nvidia.com/gpu"
triton.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	1000m
triton.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	2Gi

Scanners

Default Resources

Name	Description	Default Value
scanners.defaultResources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	2000m
scanners.defaultResources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	2Gi
scanners.defaultResources.requests.cpu	CPU reserved for the pod for scheduling.	1000m

Name	Description	Default Value
	Kubernetes guarantees at least this amount.	
scanners.defaultResources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	1Gi

Language Scanner

Name	Description	Default Value
scanners.language.replicas	Number of replicas	1

Code Scanner

Name	Description	Default Value
scanners.code.replicas	Number of replicas	1

Prompt Injection Scanner

Name	Description	Default Value
scanners.promptinjection.replicas	Number of replicas	1
scanners.promptinjection.env.CHUNK_SIZE	Number of tokens in each text segment created before sending to scanner.	"512"
scanners.promptinjection.env.CHUNK_OVERLAP_SIZE	Number of tokens shared between adjacent segments to avoid context loss across chunk boundaries.	"256"

Sensitive Scanner

Name	Description	Default Value
scanners.sensitive.replicas	Number of replicas	1
scanners.sensitive.resources.limits.cpu	Maximum CPU the pod is allowed to use. The	2000m

Name	Description	Default Value
	container cannot exceed this limit.	
scanners.sensitive.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	10Gi
scanners.sensitive.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	1000m
scanners.sensitive.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	6Gi

Toxicity Scanner

Name	Description	Default Value
scanners.toxicity.replicas	Number of replicas	1
scanners.toxicity.env.CHUNK_SIZE	Number of tokens in each text segment created before sending to scanner.	"512"
scanners.toxicity.env.CHUNK_OVERLAP_SIZE	Number of tokens shared between adjacent segments to avoid context loss across chunk boundaries.	"256"

Anonymize Scanner

Name	Description	Default Value
scanners.anonymize.replicas	Number of replicas	1
scanners.anonymize.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	2000m
scanners.anonymize.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	40Gi

Name	Description	Default Value
scanners.anonymize.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	1000m
scanners.anonymize.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	6Gi

Deanonymize Scanner

Name	Description	Default Value
scanners.deanonymize.replicas	Number of replicas	1

Custom Rule Scanner

Name	Description	Default Value
scanners.customrule.replicas	Number of replicas	1

License Manager

Name	Description	Default Value
license_manager.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	500m
license_manager.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	512Mi
license_manager.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	250m
license_manager.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	256Mi

LogD Service

Name	Description	Default Value
logd.replicas	Number of replicas	1
logd.resources.limits.cpu	Maximum CPU the pod is allowed to use. The container cannot exceed this limit.	500m
logd.resources.limits.memory	Maximum memory the pod is allowed to use. The container cannot exceed this limit.	512Mi
logd.resources.requests.cpu	CPU reserved for the pod for scheduling. Kubernetes guarantees at least this amount.	250m
logd.resources.requests.memory	Memory reserved for the pod for scheduling. Kubernetes guarantees at least this amount	256Mi

Storage

Name	Description	Default Value
storage.size	Persistent volume size	8Gi
storage.storageClass	Storage class	""
storage.existingPvName	Existing PV name	""

Ingress

Name	Description	Default Value
ingress.className	Ingress class name	"nginx"
ingress.host	Hostname for routing	""
ingress.annotations	Custom ingress annotations	{}

TLS

Name	Description	Default Value
tls.enabled	Enable TLS for services (except PostgreSQL/Redis)	true
tls.mountPath	TLS mount path in containers	"/etc/tls"
tls.cert	Certificate used by FortiAIGate TLS services	files/certificate/dflt.crt
tls.key	The key of the certificate used by FortiAIGate TLS services	files/certificate/dflt.key

PostgreSQL

Name	Description	Default Value
postgresql.enabled	Enable PostgreSQL service	true
postgresql.image.registry	Image registry	docker.io
postgresql.image.repository	Image repository	bitnamilegacy/postgresql
postgresql.image.tag	Image tag	17.4.0-debian-12-r19
postgresql.image.pullPolicy	Image pull policy	IfNotPresent
postgresql.auth.username	Database username	"fortiaigate_postgres_user"
postgresql.auth.database	Database name	"fortiaigate_db"
postgresql.tls.enabled	Enable TLS	true
postgresql.tls.certificatesSecret	Existing TLS secret	"fortiaigate-tls-secret"
postgresql.primary.persistence.size	Storage size	8Gi
postgresql.primary.resources.limits.cpu	CPU limit	1000m
postgresql.primary.resources.limits.memory	Memory limit	4Gi
postgresql.primary.service.port	Service port	5432

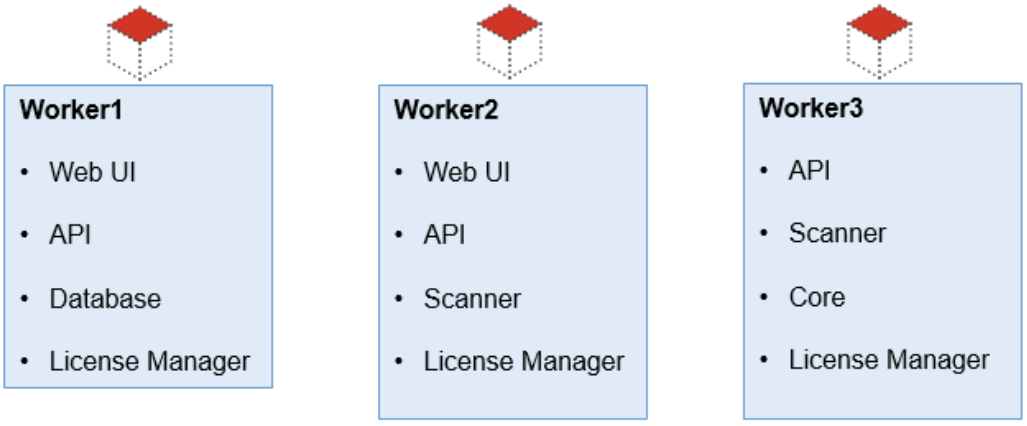
Redis

Name	Description	Default Value
redis.enabled	Enable Redis service	true
redis.image.registry	Image registry	docker.io
redis.image.repository	Image repository	bitnamilegacy/redis
redis.image.tag	Image tag	8.2.0-debian-12-r0
redis.architecture	Deployment mode	standalone
redis.tls.enabled	Enable TLS	true
redis.master.persistence.size	Storage size	4Gi
redis.master.resources.limits.cpu	CPU limit	1000m
redis.master.resources.limits.memory	Memory limit	2Gi
redis.master.service.port	Service port	6379

▼ 4. License Configuration

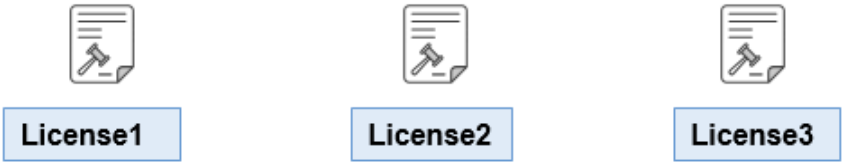
▼ 4.1 Determine which worker nodes will join the FortiAIGate cluster

Determine which worker nodes will be used by FortiAIGate and write down the names of these worker nodes, for example, Worker1, Worker2, and Worker3.



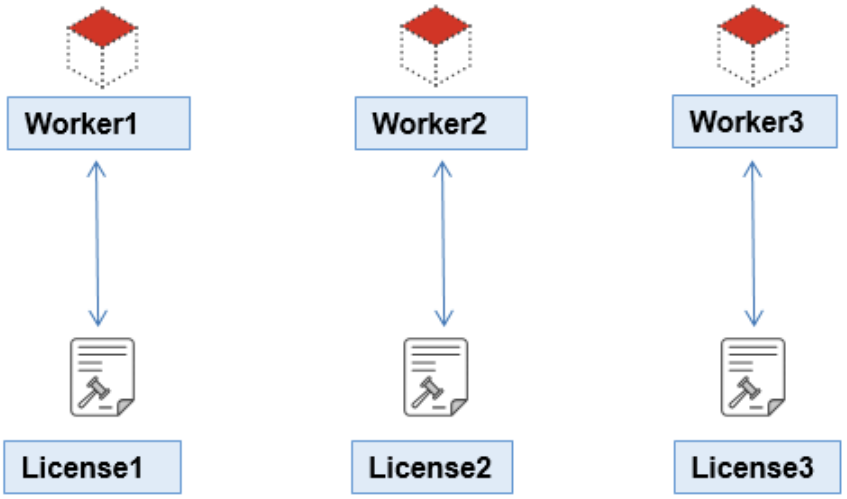
▼ 4.2 Download the license files from FortiCare

Log in to FortiCare and download the license files from the portal. Write down the file names of these licenses, for example, License1, License2, and License3.



▼ 4.3 Determine the node-license mappings

Determine which license file will be installed on which worker node by creating node-license mappings, for example, Worker1-License1, Worker2-License2, Worker3-License3.



▼ 4.4 Configure the license keys and values

1. Upload the license file to `<helm_chart_extract_path>/fortiaigate/files/licenses` directory.
2. Open the Helm chart values file (values.yaml) and configure the values under licenses. For example:

```
global:
  security:
    allowInsecureImages: true

  # License Management
  # Map of node names to their license files.
  # Pods will ONLY be scheduled on these specific nodes.
  # Example:
  # licenses:
  #   "worker-node-1": files/licenses/license-1.lic
  #   "worker-node-2": files/licenses/license-2.lic
  licenses:
    "worker1": files/licenses/license1.lic
    "worker2": files/licenses/license2.lic
    "worker3": files/licenses/license3.lic
```

Note:

- The key is the name of the Kubernetes node. FortiAIGate pods are only deployed on the nodes specified by these keys.
- Each license can only be assigned to one node

▼ 5. Deploy FortiAIGate

▼ 5.1 Create Namespace

```
kubectl create namespace fortiaigate
```

▼ 5.2 Install Helm Chart

```
cd <helm_chart_extract_path>

helm install fortiaigate ./fortiaigate \
  -n fortiaigate \
  -f ./fortiaigate/values.yaml
```

▼ 5.3 Verify Deployment

Check Pods:

```
kubectl get pods -n fortiaigate
```

All pods should be running and ready.

```
NAME                                          READY   STATUS    RESTARTS        AGE
anonymize-scanner-67448997f8-kxmw8            1/1     Running   0               4h26m
api-696cd87cdb-2h8lx                          1/1     Running   1 (4h25m ago)   4h26m
code-scanner-74d5cbb97-slx8x                  1/1     Running   0               4h26m
core-6747c9f668-hlhnp                         1/1     Running   1 (4h25m ago)   4h26m
customrule-scanner-68c848648f-jrf9h           1/1     Running   0               4h26m
deanonymize-scanner-5b859864c9-2ncd9          1/1     Running   0               4h26m
fortiaigate-postgresql-0                      1/1     Running   0               4h25m
fortiaigate-redis-master-0                    1/1     Running   0               27h
language-scanner-85484cc7d5-7hq6v             1/1     Running   0               4h26m
license-manager-4www6                         1/1     Running   0               4h26m
license-manager-nfgx5                         1/1     Running   0               4h26m
logd-b8b9c74cc-58lkb                          1/1     Running   1 (4h25m ago)   4h26m
promptinjection-scanner-59b596c88d-2ltd6      1/1     Running   0               4h26m
sensitive-scanner-84ccdf9966-fj8xm            1/1     Running   0               4h26m
toxicity-scanner-65c7bf7554-gp9ms             1/1     Running   0               4h26m
triton-server-65d7b79f8-zxhhp                 1/1     Running   0               31h
webui-bbbbb468b-cghdx                         1/1     Running   0               4h26m
```

Check Services:

```
kubectl get svc -n fortiaigate
```

```
root@k8s-master-131:/home/FortiAiGate/k8s-main-20260211# kubectl get svc -n fortiaigate
NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)             AGE
anonymize                   ClusterIP   10.103.231.198   <none>        50051/TCP           7d2h
api                         ClusterIP   10.98.145.119    <none>        8000/TCP            7d2h
code                        ClusterIP   10.109.206.201   <none>        50051/TCP           7d2h
core                        ClusterIP   10.101.147.19    <none>        8080/TCP            7d2h
customrule                  ClusterIP   10.110.60.27     <none>        50051/TCP           7d2h
deanonymize                 ClusterIP   10.109.220.223   <none>        50051/TCP           7d2h
fortiaigate-postgresql      ClusterIP   10.99.94.250     <none>        5432/TCP            7d2h
fortiaigate-postgresql-hl   ClusterIP   None             <none>        5432/TCP            7d2h
fortiaigate-redis-headless  ClusterIP   None             <none>        6379/TCP            7d2h
fortiaigate-redis-master    ClusterIP   10.101.128.133   <none>        6379/TCP            7d2h
language                    ClusterIP   10.106.178.63    <none>        50051/TCP           7d2h
license-manager             ClusterIP   10.106.5.16      <none>        9000/TCP,9443/TCP   7d2h
promptinjection             ClusterIP   10.98.22.183     <none>        50051/TCP           7d2h
sensitive                   ClusterIP   10.96.220.123    <none>        50051/TCP           7d2h
toxicity                    ClusterIP   10.104.157.116   <none>        50051/TCP           7d2h
triton-server               ClusterIP   10.102.68.181    <none>        8001/TCP,8002/TCP   7d2h
webui                       ClusterIP   10.104.200.145   <none>        3000/TCP            7d2h
root@k8s-master-131:/home/FortiAiGate/k8s-main-20260211#
```

Check Ingress:

```
kubectl get ingress -n fortiaigate
```

```
root@k8s-master-131:/home/FortiAiGate/k8s-main-20260211# kubectl get ingress -n fortiaigate
NAME                 CLASS   HOSTS   ADDRESS        PORTS     AGE
fortiaigate-ingress  nginx   *       [redacted]     80, 443   7d2h
```

```
kubectl describe ingress -n fortiaigate
```

```
Name:             fortiaigate-ingress
Labels:           app.kubernetes.io/managed-by=Helm
                  app.kubernetes.io/name=fortiaigate
                  app.kubernetes.io/part-of=fortiaigate-platform
Namespace:        fortiaigate
Address:          10.200.7.132
Ingress Class:    nginx
Default backend:  <default>
TLS:
  fortiaigate-tls-secret terminates myaigate.faigtest.com
Rules:
  Host                  Path  Backends
  ----                  ----  --------
  myaigate.faigtest.com
                        /       webui:3000 (10.250.62.237:3000)
                        /api/   api:8000 (10.250.62.204:8000)
                        /v1/    core:8080 (10.250.62.219:8080)
Annotations:            meta.helm.sh/release-name: fortiaigate
                        meta.helm.sh/release-namespace: fortiaigate
                        nginx.ingress.kubernetes.io/backend-protocol: HTTPS
                        nginx.ingress.kubernetes.io/force-ssl-redirect: true
                        nginx.ingress.kubernetes.io/healthcheck-protocol: HTTPS
                        nginx.ingress.kubernetes.io/ssl-redirect: true
Events:                 <none>
```

▼ 6. Access FortiAIGate

If using Ingress:

```
kubectl get svc -n ingress-nginx
```

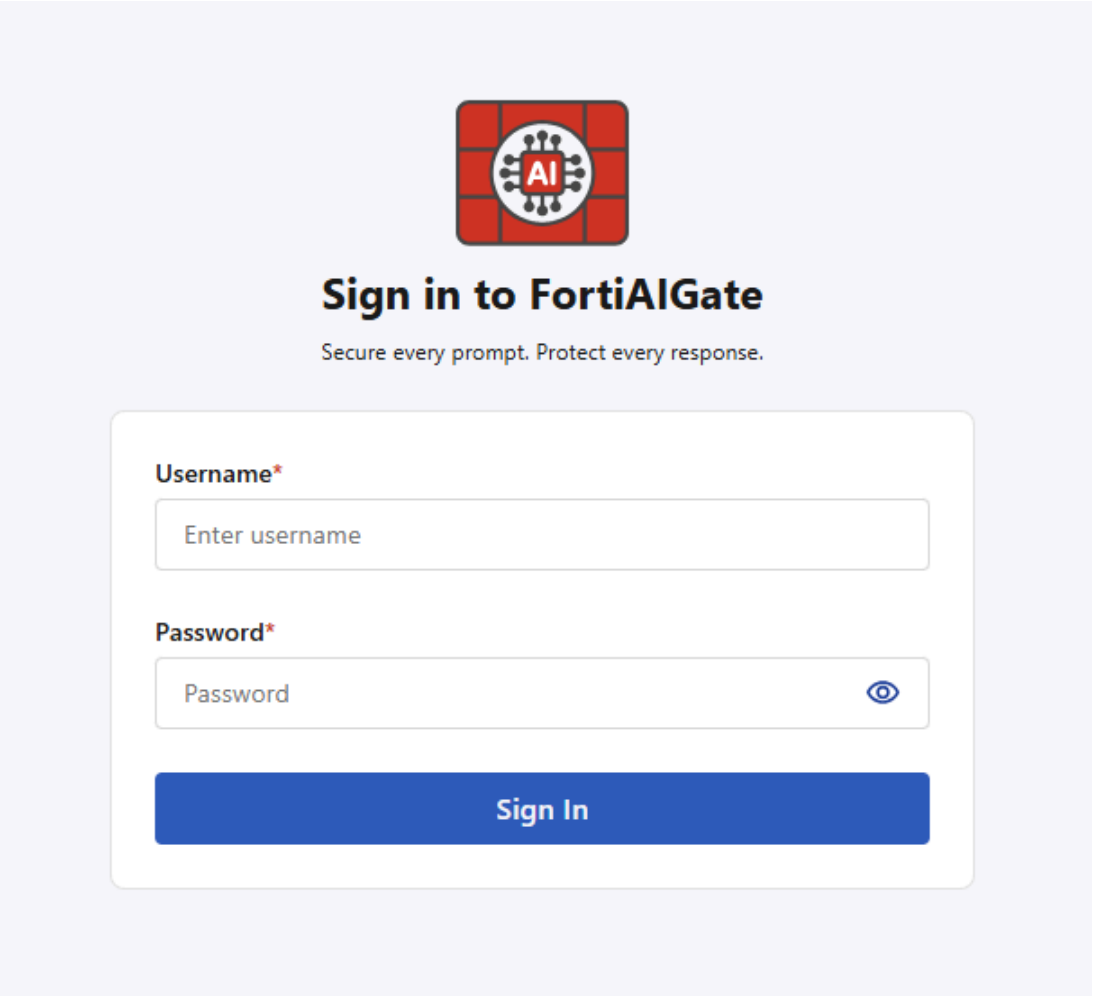
```
NAME                                 TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)                      AGE
ingress-nginx-controller             NodePort    10.101.244.140   <none>        80:30080/TCP,443:30443/TCP   22d
ingress-nginx-controller-admission   ClusterIP   10.96.243.192    <none>        443/TCP                      22d
```

Use https://<configured-domain>:<port> or https://<Node_IP>:<port>

If using LoadBalancer:

```
kubectl get svc -n fortiaigate
```

Use https://<External-IP>.



▼ 7. Upgrade FortiAIGate

If you need to update FortiAIGate, adjust the relevant settings in `values.yaml`, then run the following command to update:

```
helm upgrade fortiaigate ./fortiaigate \
  -n fortiaigate \
  -f ./fortiaigate/values.yaml
```

To upgrade FortiAIGate to a new version,

- obtain the updated images and Helm chart
- push the images to your registry server (refer to the image preparation steps),
- update the image repository and tag fields in `values.yaml` of the new Helm chart accordingly
- and then run the following upgrade command:

```
helm upgrade fortiaigate ./fortiaigate \
  -n fortiaigate \
  -f ./fortiaigate/values.yaml
```